describes studies that examine relationships between organisms and their environment across large areas to explain patterns of abundance, diversity, distribution, and change.

Brown had tried and tested this methodology in the 1970s while studying the potential effects of global warming on species in cool, moist forest and meadow habitats on 19 isolated ridges of the Great Basin, in California and Utah. He realized it would take years of fresh fieldwork to collect enough data. Instead, he used existing findings to draw new conclusions. First, he predicted how much shrinkage would occur in the area of ridge-top habitat with an assumed increase in temperature. Using known data on the minimum area required to support a population of each small mammal species, Brown was able to work out the extinction risk on each ridge as temperatures rose, and suggest conservation priorities.

Enhancing fieldwork

Macroecology often supplements fieldwork and can lead to surprising discoveries. In Madagascar, satellite data was used to develop models for chameleon species and predict them in areas beyond their known ranges. As a result, scientists investigating these areas found several new sister species.